

Part VII — Fluids · Chapter 37

Fluid Flow

How fluids move, how flow rate is described, and how continuity governs motion through varying cross-sections

Turn on a garden hose and observe: when you cover most of the opening with your thumb, the water shoots out much faster and farther than before. The volume of water leaving per second does not change — but the speed of the water increases dramatically as the opening narrows. The same effect causes rivers to run fastest in their narrowest gorges and slowest in their wide floodplains. Blood speeds up as it passes through narrowed arteries. Air accelerates through the throat of a carburetor. All of these are expressions of a single idea: because fluid cannot be created or destroyed, any decrease in cross-sectional area through which a fluid flows must be accompanied by an increase in flow speed. This chapter develops the mathematics of fluid motion, starting from the concept of flow rate and building to the continuity equation that formalises this conservation of matter in moving fluids.

37.1 What It Means for a Fluid to Flow

In the previous chapters, we treated fluids at rest (hydrostatics). We now turn to fluids in motion (fluid dynamics). A fluid flows when there is a net force driving it — typically a pressure difference. Fluid flows from regions of higher pressure toward regions of lower pressure, just as objects accelerate from regions of higher to lower potential energy.

Flow can be described at two levels: macroscopic and microscopic. At the macroscopic level, we treat the fluid as a continuous medium and describe properties like velocity, pressure, and density as functions of position and time. This is the continuum assumption, valid whenever the fluid parcel under consideration contains enormous numbers of molecules (essentially always true for liquids, and true for gases at ordinary pressures). At the microscopic level, individual molecules are tracked — this level of detail is only needed for rarefied gases or very small-scale flows (microfluidics).

In this chapter we adopt the macroscopic view and further simplify by considering ideal fluid flow: the fluid is incompressible (density is constant, valid for liquids and slow gas flows), inviscid (frictionless — no internal friction or viscosity), and the flow is

steady (conditions at every point do not change with time). These idealisations allow clean mathematical results; the deviations from ideal behaviour (viscosity, compressibility, turbulence) are discussed at the end of the chapter.

Key Point: Ideal fluid flow is a useful approximation for: water in pipes at moderate speeds, air around slowly moving objects, and preliminary design calculations. Corrections for viscosity (internal friction) and turbulence become important at high flow speeds or in narrow channels.

37.2 Flow Rate and Volume Flow Rate

Volume flow rate (also called volumetric flow rate) Q is the volume of fluid passing a cross-section per unit time. It is the most natural description of “how fast” a fluid is moving through a pipe, channel, or duct:

$$Q = V / t$$

Volume flow rate Q (m^3/s) = volume V (m^3) $\div$ time t (s). Other common units: litres per second (L/s), litres per minute (L/min), cubic centimetres per second (cm^3/s). $1 \text{ L/s} = 10^{-3} \text{ m}^3/\text{s}$.

Flow rate can also be expressed in terms of the flow speed v and the cross-sectional area A perpendicular to the flow direction. In time Δt , the fluid moves a distance $v\Delta t$. The volume that passes the cross-section in this time is:

$$Q = A v$$

Q = cross-sectional area A (m^2) $\times$ average flow speed v (m/s). This form is fundamental: it connects the geometric description (area) to the kinematic description (speed) of flow.

Example — Flow rate in a pipe Water flows through a circular pipe of diameter 4.0 cm at a speed of 2.5 m/s. Area = $\pi(0.02)^2 = 1.257 \times 10^{-3} \text{ m}^2$. $Q = Av = 1.257 \times 10^{-3} \times 2.5 = 3.14 \times 10^{-3} \text{ m}^3/\text{s} = 3.14 \text{ L/s}$.

Example — Speed from flow rate A pump delivers water at $Q = 60 \text{ L/min} = 10^{-3} \text{ m}^3/\text{s}$ through a pipe of radius 1.5 cm. $v = Q/A = 10^{-3} / (\pi \times 0.015^2) = 10^{-3} / (7.07 \times 10^{-4}) = 1.41 \text{ m/s}$.

37.3 Mass Flow Rate (Conceptual Introduction)

Mass flow rate $\dot{m}$ (m-dot) is the mass of fluid passing a cross-section per unit time. It is related to volume flow rate by the fluid density:

$$\dot{m} = \rho Q = \rho A v$$

Mass flow rate $\dot{m}$ (kg/s) = density ρ (kg/m³) $\times$ volume flow rate Q (m³/s). For incompressible fluids, ρ is constant, so mass flow rate and volume flow rate carry the same information.

Mass flow rate is important in thermodynamics (energy carried by flowing fluids), engineering (combustion and reaction rates, which depend on mass not volume), and anywhere the fluid density changes significantly. For the incompressible liquids considered in this chapter, we can work with volume flow rate Q and convert to mass flow rate when needed.

For a compressible gas (not treated in this chapter), density changes with pressure and temperature, so volume flow rate alone is ambiguous — you must specify pressure and temperature. Mass flow rate is unambiguous because it directly counts molecules. This is why gas flow is often specified in kg/s or standard cubic metres per second (where “standard” refers to a defined reference pressure and temperature).

37.4 Streamlines and Flow Paths

A streamline is a curve that is everywhere tangent to the velocity vector of the fluid at a given instant. In steady flow, streamlines are also the paths that individual fluid elements follow: a particle released into the flow at a given point will travel along the streamline through that point.

Streamlines have several important properties. They never cross each other in steady flow: if two streamlines crossed, a fluid element at the crossing point would have two different velocities simultaneously, which is impossible. Streamlines are denser (closer together) where the flow speed is higher and spaced farther apart where the flow speed is lower. This visual property of streamline patterns is extremely useful for qualitative analysis.

A stream tube is a bundle of streamlines forming a tube-like region. No fluid crosses the walls of a stream tube (by definition, since streamlines are parallel to velocity). Fluid entering one end of a stream tube must exit the other end. This concept is the geometric basis for the continuity equation.

Key Point: In unsteady (time-varying) flow, streamlines change from moment to moment and no longer coincide with particle paths. In that case, particle paths are called pathlines, and are distinct from streamlines. For steady flow only, streamlines =

pathlines = streaklines (the line traced by dye continuously injected at a point).

37.5 Steady vs. Unsteady Flow

Flow is steady if the velocity, pressure, and density at every fixed point in the flow do not change with time. A tap running at constant rate produces approximately steady flow in the pipe. Flow is unsteady if these properties change with time at fixed points — as when a tap is turned on or off, or when waves propagate through a fluid.

Most practical pipe and duct flows reach a quasi-steady state a short time after startup. The transient period (when flow is establishing itself) involves unsteady effects. Once steady state is reached, the simpler steady-flow equations apply.

Natural flows are more often unsteady: the heartbeat creates pulsatile (periodic unsteady) blood flow; wind is inherently unsteady; ocean waves are unsteady surface waves. The steady-flow assumption is an idealisation that gives useful results when the time scale of flow changes is much longer than the transit time of fluid through the system.

37.6 Laminar vs. Turbulent Flow

Laminar flow (also called streamline flow or viscous flow) occurs when fluid layers slide smoothly past each other in parallel paths, without lateral mixing. The streamlines are smooth, orderly, and nearly parallel. Laminar flow occurs at low speeds in narrow channels or with viscous fluids.

Turbulent flow occurs at high speeds or in wide channels. The flow breaks into chaotic eddies and swirls; fluid elements mix laterally and the velocity at any point fluctuates irregularly with time. Turbulent flow carries much greater energy in its random motions, produces greater friction losses, and is much harder to model mathematically.

The transition between laminar and turbulent flow is predicted by the Reynolds number Re , a dimensionless ratio of inertial to viscous forces:

$$Re = \rho v D / \mu$$

Reynolds number Re (dimensionless). ρ = fluid density (kg/m^3); v = flow speed (m/s); D = characteristic length (pipe diameter for pipe flow, m); μ = dynamic viscosity ($\text{Pa}\cdot\text{s}$). For pipe flow: $Re < 2300 \rightarrow$ laminar; $Re > 4000 \rightarrow$ turbulent; $2300\text{--}4000 \rightarrow$ transitional.

Flow Type	Re (pipe)	Characteristics	Examples
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Laminar	< 2300	Smooth, ordered layers; no lateral mixing; parabolic velocity profile	Honey flowing slowly; blood in capillaries; viscous oil in tubes
Transitional	2300–4000	Intermittent; switches between laminar and turbulent	Water tap at moderate flow
Turbulent	> 4000	Chaotic eddies; strong lateral mixing; flatter velocity profile	Water from a fully open tap; river rapids; wind at ordinary speeds

Table 37.1 *Classification of pipe flow by Reynolds number.*

For this chapter and Chapter 38, we assume laminar, steady, ideal flow. Real pipe flow is turbulent in most engineering applications ($Re \gg 4000$), but the ideal flow equations still give correct order-of-magnitude results and the right physical intuition about the relationship between speed and area.

37.7 Continuity in Fluid Flow

Continuity is the principle that matter is conserved in fluid flow. No fluid is created or destroyed inside a flow system (no sources or sinks). This means that whatever mass enters a pipe or channel per unit time must also leave it per unit time.

Consider fluid flowing through a pipe that changes cross-sectional area from A_1 at position 1 to A_2 at position 2. In a small time interval Δt , the fluid at position 1 moves a distance $v_1 \Delta t$, sweeping out a volume $A_1 v_1 \Delta t$. The mass entering the wide section in this time is $\rho A_1 v_1 \Delta t$. The mass leaving the narrow section is $\rho A_2 v_2 \Delta t$. For an incompressible fluid with no accumulation:

$$\rho A_1 v_1 = \rho A_2 v_2$$

Since ρ is constant for an incompressible fluid, it cancels, giving the continuity equation: $A_1 v_1 = A_2 v_2$, or equivalently $Q_1 = Q_2$.

37.8 The Continuity Equation

The Rule: Continuity Equation: $A_1 v_1 = A_2 v_2$ (for steady, incompressible flow). The

volume flow rate $Q = Av$ is constant throughout any flow path with no branches or merges.

$$A_1 v_1 = A_2 v_2 \quad \Rightarrow \quad Q_1 = Q_2$$

The continuity equation. Where cross-sectional area is smaller, flow speed is greater; where area is larger, flow speed is smaller. Q (m^3/s) is constant along any unbranched streamtube.

The continuity equation is the fluid equivalent of conservation of mass (or, for incompressible fluids, conservation of volume). It constrains the kinematics of the flow — the speed at any point is determined by the geometry (area) of the flow path at that point, given the flow rate Q .

When a pipe branches into two or more branches, continuity applies to the total flow: the flow rate entering equals the sum of flow rates leaving. If a pipe of area A_0 splits into two branches of areas A_1 and A_2 :

$$Q_0 = Q_1 + Q_2 \quad \Rightarrow \quad A_0 v_0 = A_1 v_1 + A_2 v_2$$

Continuity at a branch point. Used in pipe networks, cardiovascular physiology, river systems, and electrical circuit analogies.

Example — Continuity in a narrowing pipe Water flows through a pipe that narrows from diameter 8.0 cm to 3.0 cm. In the wide section, $v_1 = 0.50$ m/s. Find the speed in the narrow section. $A_1 = \pi(0.04)^2 = 5.03 \times 10^{-3} \text{ m}^2$; $A_2 = \pi(0.015)^2 = 7.07 \times 10^{-4} \text{ m}^2$. $v_2 = A_1 v_1 / A_2 = (5.03 \times 10^{-3} \times 0.50) / (7.07 \times 10^{-4}) = 3.56$ m/s. The speed increases by a factor of $A_1/A_2 = (8/3)^2 = 7.11$.

37.9 Flow Through Pipes and Constrictions

The continuity equation directly explains several observable phenomena in everyday pipe flow. When you partially block a garden hose nozzle with your thumb, you reduce the cross-sectional area A . Since Q is fixed by the pump or water mains pressure, $v = Q/A$ must increase — the water sprays farther. The same principle governs spray nozzles, showerheads, and firefighting hoses.

In a river, water flows slowly in wide, shallow sections (floodplains) and rapidly in narrow gorges. The volume flow rate (discharge) of the river is the same throughout; the product Av is constant. The narrower the channel (smaller A), the faster the water (larger v).

In the human cardiovascular system, the aorta (large artery, diameter ~ 2.5 cm) carries blood away from the heart. Blood then branches into progressively smaller arteries, arterioles, and capillaries. Although each individual capillary is tiny (diameter ~ 8 μm), there are so many capillaries in parallel that their total cross-sectional area is vastly

greater than the aorta. Continuity therefore requires blood to flow much more slowly in the capillaries than in the aorta — which is physiologically important because the slow capillary flow allows time for oxygen and nutrient exchange with surrounding tissues.

Example — Blood flow and continuity The aorta has cross-sectional area 4.5 cm^2 ; blood flows at 30 cm/s . $Q = 4.5 \times 30 = 135 \text{ cm}^3/\text{s} = 135 \text{ mL/s} \approx 8.1 \text{ L/min}$ (normal cardiac output). In the capillaries, total cross-sectional area $\approx 4500 \text{ cm}^2$ (10^3 times larger). By continuity, capillary blood speed $= 135/4500 \approx 0.03 \text{ cm/s} = 0.3 \text{ mm/s}$. This is about 1000 times slower than in the aorta.

37.10 Real-World Flow Systems

The ideal pipe flow model (steady, incompressible, inviscid) gives useful intuition, but real engineering flow systems involve additional considerations. Four of the most important are: pipe friction losses, entrance and exit effects, flow measurement, and system branching.

Pipe friction: Real fluids are viscous. Viscosity causes friction between the fluid and pipe walls, which dissipates energy and causes a pressure drop along the pipe. The Darcy-Weisbach equation (beyond this chapter) quantifies this pressure drop as a function of flow speed, pipe diameter, length, and friction factor. In general, higher speed flow experiences greater friction losses.

Flow measurement: Several practical devices exploit the continuity equation and pressure changes to measure flow rate. The Venturi meter constricts the pipe to speed up the flow, then measures the pressure difference between the wide and narrow sections. The orifice plate does the same with a simpler (but less efficient) constriction. The Pitot tube measures local flow speed from the pressure difference between a forward-facing orifice (stagnation pressure) and a side-facing orifice (static pressure). All of these are discussed further in Chapter 38 under Bernoulli applications.

System branching: Municipal water distribution networks, natural gas grids, and biological vascular systems all involve complex branching pipe networks. Continuity applies at every junction, and pressure must be consistent at each node. Solving such networks requires simultaneous application of continuity (flow conservation at junctions) and pressure equations (energy conservation along paths).

37.11 Limits of Ideal Flow Models

The ideal flow model (incompressible, inviscid, steady) is a powerful tool but has well-

defined limits. Understanding these limits helps identify when more sophisticated models are needed.

Compressibility becomes important when fluid speeds approach the speed of sound. For air, this occurs at Mach 0.3 or above (about 100 m/s at sea level). Aircraft flying at Mach 0.8–1.2 (transonic) or above (supersonic) cannot use incompressible flow equations. Shockwaves form when local speeds exceed the sound speed, creating sudden pressure jumps that the ideal model cannot predict.

Viscosity is always present in real fluids. Its effects are negligible at high Reynolds numbers (turbulent flows in wide pipes) for most macroscopic purposes, but dominate at low Reynolds numbers (microfluidics, creeping flow). The no-slip condition — that the fluid velocity at a solid wall must equal the wall velocity (zero for stationary walls) — is a viscous effect that the ideal model ignores. In reality, a thin boundary layer near every wall has a velocity gradient from zero (at the wall) to the free-stream value. Boundary layer theory is a major subject in fluid mechanics.

Turbulence: Even when the bulk flow is well modelled by the steady equations, local instabilities can trigger turbulent fluctuations. Turbulence dramatically increases effective friction and mixing but is extremely difficult to model from first principles. Most engineering design uses empirical correlations for turbulent friction, established by experiments.

Bridge Note *The continuity equation derived in this chapter is the kinematic statement of conservation of mass. Chapter 38 introduces Bernoulli's equation, the energetic statement — conservation of energy in ideal flow. Together, continuity and Bernoulli allow analysis of a wide range of fluid systems.*

37.12 Common Mistakes in Fluid Flow Analysis

- Using diameter instead of radius (or area) in $Q = Av$. Area = $\pi r^2 = \pi(d/2)^2$. Using d directly instead of $d/2$ gives an area four times too large.
- Forgetting to square the radius when computing circular cross-sectional area. If r doubles, A quadruples ($A \propto r^2$). Students often apply a linear ratio where a squared ratio is needed.
- Applying continuity to the wrong cross-section. Continuity $A_1v_1 = A_2v_2$ applies between any two points in the same unbranched streamtube. If there is a branch between the two points, total flow conservation must be used instead.
- Confusing flow rate Q with flow speed v . $Q = Av$ has units m^3/s ; v has units m/s . A garden hose and a river may have the same flow speed but very different flow

rates (because of different areas). Or the same flow rate but very different speeds.

- Assuming that a larger pipe always means faster flow. By continuity, a larger area pipe has slower speed at the same flow rate. The water main in your street has a much larger diameter than your household pipes and flows much more slowly, even though it supplies a higher total flow rate.
- Applying ideal flow (continuity only) to flows where viscosity matters significantly. In very narrow tubes (capillaries, microchannels) or with very viscous fluids (oil, honey), viscous effects dominate and the inviscid continuity equation gives incorrect velocities.

Common Mistake: When using the continuity equation, always compute area from cross-sectional dimensions: $A = \pi r^2$ for circular pipes; $A = \text{width} \times \text{depth}$ for rectangular channels. Never use diameter directly as area, and never confuse the ratio of areas with the ratio of diameters.

Chapter Summary

Fluid flow is described by velocity fields, flow rate, and streamlines. In ideal (steady, incompressible, inviscid) flow, the continuity equation expresses conservation of mass.

- Volume flow rate: $Q = Av$ (m^3/s). Mass flow rate: $\dot{m} = \rho Q = \rho Av$ (kg/s).
- Streamlines are everywhere tangent to velocity; they never cross in steady flow.
- Laminar flow is smooth and ordered; turbulent flow is chaotic. The Reynolds number $Re = \rho v D / \mu$ predicts the transition.
- Continuity equation: $A_1 v_1 = A_2 v_2$ (Q constant along an unbranched streamtube). At a branch: $Q_{\text{in}} = \Sigma Q_{\text{out}}$.
- Smaller cross-section $\rightarrow$ faster flow; larger cross-section $\rightarrow$ slower flow (at constant Q).
- Applications: pipe networks, cardiovascular flow, river channels, nozzles, flow meters.
- Ideal flow limits: compressibility important at $Mach > 0.3$; viscosity important at low Re ; turbulence requires empirical models.

Key Terms

fluid dynamics: The study of fluids in motion, as opposed to fluid statics (fluids at rest).

volume flow rate (Q): $Q = Av$; the volume of fluid passing a cross-section per unit time (m^3/s). Also called volumetric flow rate or discharge.

mass flow rate ($\dot{m}$): $\dot{m} = \rho Q = \rho Av$; the mass of fluid passing a cross-section per unit time (kg/s).

streamline: A curve everywhere tangent to the fluid velocity field; in steady flow, the path followed by a fluid element.

stream tube: A bundle of streamlines forming a tube through which fluid flows; no fluid crosses the tube walls.

steady flow: Flow in which velocity, pressure, and density at every fixed point do not change with time.

laminar flow: Smooth, ordered flow in parallel layers with no lateral mixing; occurs at low Reynolds numbers.

turbulent flow: Chaotic, irregular flow with strong lateral mixing and eddies; occurs at high Reynolds numbers.

Reynolds number (Re): $Re = \rho v D / \mu$; dimensionless ratio of inertial to viscous forces; predicts laminar-turbulent transition.

continuity equation: $A_1 v_1 = A_2 v_2$ (or $Q_1 = Q_2$); the mathematical expression of mass conservation for steady, incompressible flow.

ideal fluid: A hypothetical fluid that is incompressible and inviscid (frictionless); used to derive the continuity and Bernoulli equations.

no-slip condition: The physical requirement that real fluid velocity equals wall velocity at any solid boundary (zero for a stationary wall); a viscous effect absent in ideal flow.

Concept Review Questions

Answer in complete sentences. No calculations required unless stated.

1. Explain in terms of flow rate and continuity why water shoots farther when you partially block a garden hose nozzle with your thumb.
2. A river flows from a mountain gorge into a wide valley. In which location is the water moving faster? Explain using the continuity equation.
3. Define streamline. Why can two streamlines never cross each other in steady flow?

4. Distinguish laminar flow from turbulent flow. What quantity predicts which type of flow occurs in a pipe?
 5. Write the continuity equation for a branching pipe where one pipe of area A_0 splits into three pipes of areas A_1 , A_2 , and A_3 . What does it express physically?
 6. Blood in the human body moves slowly in capillaries but faster in the aorta. Explain this using the continuity equation and the fact that capillaries have enormous total cross-sectional area.
 7. A pipe of constant diameter has a flow speed of 3 m/s throughout. Does the continuity equation tell us anything useful here? What does it confirm?
 8. Identify two physical effects that cause real pipe flow to deviate from ideal (inviscid, incompressible) flow. For each, state the conditions under which the ideal model fails most seriously.
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Flow Rate Problems

9. A fire hose (diameter 6.0 cm) delivers water at a speed of 4.0 m/s. (a) Find the cross-sectional area of the hose. (b) Find the volume flow rate in m^3/s and L/s. (c) How many litres are delivered in 2 minutes?
 10. A garden hose has an inside diameter of 1.6 cm and delivers 12 L/min. (a) Find the flow rate in m^3/s . (b) Find the flow speed in the hose. (c) The nozzle narrows to a diameter of 0.4 cm. Find the speed at the nozzle outlet.
 11. A river is 40 m wide and 2.5 m deep on average. The water flows at an average speed of 0.8 m/s. Find the discharge (volume flow rate) of the river in m^3/s and in megalitres per day (1 ML = 10^6 L).
 12. A pump fills a 500 L water tank in 4.0 minutes. (a) Find the volume flow rate in L/s and m^3/s . (b) The pump outlet has a diameter of 2.5 cm. Find the outlet flow speed.
 13. A blood vessel (aorta) of diameter 2.5 cm carries blood at 25 cm/s. Find the cardiac output (Q) in mL/s and L/min.
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Continuity Equation Exercises

14. Water flows through a pipe that widens from a diameter of 3.0 cm to 9.0 cm. The speed in the narrow section is 6.0 m/s. Find the speed in the wide section.
15. A circular pipe of radius 4.0 cm carries water at 2.0 m/s. It narrows to radius 1.5 cm. (a) Find the volume flow rate. (b) Find the speed in the narrow section.
16. A rectangular irrigation channel is 1.2 m wide and 0.40 m deep, with water flowing at 0.6 m/s. Downstream, the channel narrows to 0.80 m wide but the same depth. Find the speed in the narrow section.
17. A main water pipe (diameter 20 cm, speed 0.5 m/s) branches into four equal pipes. The water speed in each branch is 2.0 m/s. Find the diameter of each branch pipe.
18. The aorta (area 4.5 cm^2 , speed 30 cm/s) branches into two common iliac arteries of equal size. If blood speed in each iliac artery is 20 cm/s, find the area and diameter of each artery.

Pipe Flow Analysis

19. A household plumbing system has a main pipe (diameter 25 mm, speed 1.2 m/s) that feeds three branch lines of diameter 12 mm each. Assume equal flow distribution to each branch. Find the speed in each branch pipe. Is continuity satisfied?
20. A venturi flowmeter has an inlet diameter of 100 mm and a throat diameter of 50 mm. Water enters at 2.0 m/s. (a) Find the inlet flow rate. (b) Find the throat speed. (c) Explain why the throat speed can be used to infer flow rate even if Q is unknown.
21. A wind tunnel has a test section of area 0.5 m^2 and a contraction section of area 2.0 m^2 . Air enters the contraction at 5.0 m/s. Treat air as incompressible. (a) Find the air speed in the test section. (b) Find the flow rate through the tunnel.
22. An intravenous drip bag is connected to a patient via tubing of internal diameter 3.0 mm. The drip rate is 60 drops/minute. If each drop has volume 0.05 mL, find (a) the volume flow rate in mL/s and m^3/s , and (b) the average speed of fluid in the tube.

Applied Fluid Systems Problems

23. A municipal water main (diameter 300 mm) supplies a neighbourhood at 0.8 m/s. The flow splits into 20 equal household service pipes. Each service pipe has a diameter of 20 mm. Find the flow speed in each service pipe, and check that continuity is satisfied for the network.
24. A river discharges $250 \text{ m}^3/\text{s}$ through a gorge that is 15 m wide and 8.0 m deep. What is the average water speed through the gorge? Downstream the river widens to 200 m and is 2.5 m deep. What is the average speed in the wide section?
25. An air-conditioning duct (rectangular cross-section $0.60 \text{ m} \times 0.30 \text{ m}$) carries air at 3.0 m/s. It branches into six ceiling diffusers each of circular cross-section diameter 15 cm. Treat air as incompressible. Find the flow speed at each diffuser outlet.
26. In a simplified model of the human lung, the trachea (diameter 18 mm) branches into two bronchi (diameter 12 mm each), which each branch into 3 smaller bronchioles (diameter 5.0 mm each). If air flows through the trachea at 0.4 m/s during inhalation, find the speed in (a) each bronchus and (b) each bronchiole. Use continuity at each branching point.

Chapter 37 Checkpoint

Before moving on to Chapter 38, confirm you can do each of the following.

I can...	See Section
Define volume flow rate Q and calculate $Q = Av$	37.2
Define mass flow rate and relate it to volume flow rate and density	37.3
Define a streamline and describe its properties in steady flow	37.4
Distinguish steady from unsteady flow	37.5
Distinguish laminar from turbulent flow and state what predicts the transition	37.6
State the physical basis of the continuity equation (conservation of mass)	37.7
Write and apply the continuity equation $A_1v_1 = A_2v_2$	37.8
Apply continuity at a branching point $Q_{\text{in}} = \Sigma Q_{\text{out}}$	37.8
Explain flow behaviour in narrowing pipes, rivers, and biological vessels	37.9
Describe real flow effects that violate ideal flow assumptions	37.11
Avoid common errors (area vs diameter, branching, Q vs v)	37.12

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